

Localized Multi-Agent Debate in Small Language Models: A Judge-Proposer-Critic Framework

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Abstract. This paper presents a technical report on a localized, decentralized multi-agent debate (MAD) system designed to maximize the reasoning efficacy of Small Language Models (SLMs) with fewer than 10 billion parameters. The system deploys three interacting 7B-parameter agents in a structured Judge-Proposer-Critic topology to investigate whether an emergent consensus generated by multiple SLMs can statistically outperform the zero-shot reasoning of a single 70B-parameter frontier model. The framework integrates ReAct tool-use paradigms and Active Retrieval-Augmented Generation (RAG) for factual grounding, and is evaluated on the LegalBench-RAG benchmark. Targeted mitigation strategies are proposed to address Self-Preferential Bias in the Judge agent, hallucination loops stemming from ReAct syntax failure in SLMs, and the complexity of evaluating dynamic retrieval pipelines.

Keywords: Small Language Models · Multi-Agent Debate · Retrieval-Augmented Generation · ReAct · Legal Reasoning · Constrained Decoding

1 Research Goals

The primary goal of this research project is to maximize the reasoning efficacy of Small Language Models (SLMs)—specifically those under 10 billion parameters, such as 7B architectures—by deploying them in a highly structured, interactive debate framework. Prior work has demonstrated that multi-agent orchestration can substantially enhance SLM capabilities, with recent systems achieving competitive performance against much larger models through collaborative deliberation [6,1].

The study aims to quantitatively determine whether a synergistic *Emergent Consensus* generated by three interacting 7B models can statistically outperform the zero-shot reasoning capabilities of a single 70B-parameter frontier model. This hypothesis is grounded in evidence that multi-agent SLM systems can potentially match or exceed LLM-level performance through collaborative design [1,8]. Furthermore, the research seeks to empirically evaluate the compounding effects of integrating ReAct tool-use paradigms and Active Retrieval-Augmented Generation (RAG) as factual grounding mechanisms within this multi-agent setting.

2 Project Description

The project proposes a localized, decentralized artificial intelligence orchestration designed for a resource-constrained environment—specifically, a dual-T4 GPU setup providing approximately 32GB of total VRAM, as available on platforms such as Kaggle. It circumvents the financial and privacy costs associated with cloud-based LLM APIs by utilizing aggressively quantized models (e.g., 4-bit GGUF or AWQ formats) and efficient execution backends such as vLLM.

The core architecture is a **Judge-Proposer-Critic** multi-agent debate. The Proposer initiates an argument or solution, the Critic adversarially attacks the logic or retrieves counter-evidence, and a single Judge model evaluates the full debate trajectory to declare a final verdict. This design is motivated by the empirical success of structured debate settings in NLP tasks: the MAD framework introduced by Liang et al. [4] showed that a “tit-for-tat” argumentation process between agents—managed by a judge—leads to more reliable solutions than single-model self-reflection. More recently, multi-agent debate has been applied to tasks as diverse as fake news detection [5] and domain-specific RAG chunking [9], confirming the generality of the paradigm.

The system operates on a zero-shot/few-shot basis, relying on strict prompt engineering, constrained token sampling, and dynamic tool orchestration, without any model fine-tuning.

3 Methodology

Because the models will not undergo fine-tuning, the methodology relies heavily on inference-time orchestration and strict execution protocols.

Zero-Shot/Few-Shot Prompt Injection. Agents are assigned strict personas via system prompts, utilizing 3–5 “golden” trajectory examples to enforce the IRAC (Issue, Rule, Application, Conclusion) legal reasoning format. This few-shot strategy follows the design principles validated in SLM function-calling studies, where structured prompt injection is critical for output quality [3].

Sequential Orchestration. The debate follows a sequential pipeline managed by an orchestrator (e.g., LangGraph). The Proposer generates a structured response, which is passed to the Critic. The sequential model mirrors debate flow designs used in multi-agent systems for creative and reasoning tasks [7,8].

Constrained Sampling. To enforce structural fidelity during tool use, token generation is strictly bound by context-free grammar parsers (e.g., Guidance or Outlines), ensuring syntactically valid JSON/YAML outputs. This constraint is directly motivated by empirical findings that SLMs in zero-shot settings achieve JSON-parsable success rates below 10% [3], making compiler-level enforcement a necessity rather than an optimisation.

Mathematical Adaptive Stopping. To prevent the debate from running indefinitely or devolving into unproductive consensus, the orchestrator employs a time-varying Beta-Binomial mixture model to monitor opinion stability, triggering a halt when the statistical distribution of agent stances stabilises. This mechanism is inspired by the finding that the “adaptive break of debate” is a critical hyperparameter for MAD performance [4].

4 Literature Review

The theoretical foundation of this project intersects multi-agent dynamics, retrieval mechanisms, and constrained text generation.

4.1 Multi-Agent Debate and Emergent Consensus

MAD frameworks have been proven to mitigate the *Degeneration of Thought* (DoT) problem, in which a single model becomes overconfident in a flawed reasoning path and fails to self-correct through reflection [4]. When properly orchestrated, a debate-driven architecture encourages divergent thinking and leads to consensus outputs that carry synergistic predictive information exceeding the capacity of any individual agent.

Recent empirical work confirms the breadth of this effect. In the domain of misinformation detection, the TruEDebate (TED) system demonstrated that a structured multi-agent debate process—comprising opening statements, cross-examination, rebuttal, and closing statements—substantially surpasses traditional classification baselines in both accuracy and interpretability [5]. For text chunking in domain-specific RAG, QChunker employs a four-agent debate pipeline (question outline generator, text segmenter, integrity reviewer, and knowledge completer) to produce semantically coherent chunks that improve downstream retrieval quality [9].

For the specific challenge of using SLMs as judges, JudgeBoard proposes the Multi-Agent Judging (MAJ) framework, in which multiple SLMs with distinct reasoning profiles collaborate to approximate LLM-level judgment accuracy. Notably, MAJ using smaller backbone models performs comparably to, or better than, their larger counterparts on the MATH benchmark [1]. Complementarily, Wei et al. [7] demonstrate that a principle-guided LLM-as-a-Judge with multi-agent reward shaping can unlock creative capabilities in 7B-parameter SLMs, illustrating that judgment and generation quality are jointly improvable through multi-agent feedback.

4.2 SLM Orchestration and Structural Fragility

Beyond debate, recent work addresses how multiple SLMs can be systematically orchestrated. SLM-MUX [6] introduces a three-stage approach—model selection, architecture design, and test-time scaling—to coordinate complementary SLMs,

achieving up to 13.4% improvement on MATH over existing orchestration methods. This demonstrates that the complementarity of different small models is a resource that structured orchestration can exploit effectively.

ResilioMate [8] offers a resilient multi-agent framework that addresses both task-decomposition and adversarial robustness in SLMs, using dual-scale memory systems and collaborative bias-prevention strategies. Its 7B LeptoConnect backbone achieves a database operations score of 41.3, outperforming GPT-4’s 32.0, via collaborative cognitive load allocation.

At the same time, empirical studies such as *Small Models, Big Tasks* [3] demonstrate that SLMs operating in zero-shot settings struggle immensely with function calling, frequently achieving JSON-parsable success rates below 10%. This highlights the absolute necessity of token-level constrained sampling to ensure loop stability.

4.3 Active vs. Passive Retrieval-Augmented Generation

Traditional (Passive) RAG retrieves information once before generation begins, leaving the system prone to context bloat and noise distraction if the initial retrieval is poorly calibrated. Advanced frameworks introduce *Active RAG*, in which models iteratively anticipate future content needs and retrieve documents on demand based on low-confidence tokens.

Decoupling the search agent from the generator and training it with a *Gain Beyond RAG* (GBR) reward—quantifying the improvement in generation accuracy over a naive RAG baseline—has shown superior efficiency in multi-turn environments. The s^3 framework [2] demonstrates that a lightweight, model-agnostic searcher trained with this GBR objective requires only 2,400 training samples to consistently outperform baselines trained on $70\times$ more data, across both general and medical QA benchmarks. The multi-agent architecture proposed in this project adopts this GBR principle, assigning the Active RAG role adversarially to both Proposer and Critic agents.

5 Dataset: LegalBench-RAG

The project will be evaluated using the **LegalBench-RAG** dataset, the first benchmark specifically designed to evaluate the precise retrieval step of RAG pipelines within the legal domain. It consists of 6,858 human-annotated query-answer pairs spanning a corpus of over 79 million characters, covering diverse legal documents including NDAs, M&A agreements, commercial contracts, and privacy policies.

Unlike generative-focused benchmarks, LegalBench-RAG focuses on extracting minimal, highly relevant text segments rather than large, imprecise chunks. This prevents context window bloat, reduces inference latency, and mitigates the risk of models losing track of the prompt during extended legal reasoning—a concern directly analogous to the semantic fragmentation challenge addressed in domain-specific RAG systems [9].

Table 1. Comparison of reasoning configurations to be benchmarked.

Configuration	Retrieval	Reasoning	Strengths & Vulnerabilities
ReAct Only	None (parametric memory)	Reason + Act interleaved	High speed; severe hallucination risk on complex legal statutes due to lack of grounding.
Naive RAG	Passive, single-pass semantic search	Standard generation on prepended context	Prone to context bloat and noise distraction if initial retrieval is poorly calibrated.
Active RAG	Dynamic, multi-pass retrieval on uncertainty	Self-reflective generation	Resource-intensive; requires careful calibration to prevent over- or under-searching [2].
Hybrid (Proposed)	Active RAG used adversarially by Proposer/Critic	Multi-Agent Debate with ReAct structure [4]	Maximises grounding and peer-review; susceptible to latency bottlenecks and sequential context bloat.

6 Comparative Analysis

The project will benchmark the multi-agent system against four distinct reasoning configurations. Each configuration represents a distinct point in the design space along two axes: retrieval mechanism and reasoning paradigm. Table 1 summarises the configurations with their respective strengths and vulnerabilities.

7 Highlighted Issues and Mitigation Strategies

7.1 Self-Preferential Bias in the Judge Agent

The Issue. When a single LLM acts as an adjudicator, it exhibits a pronounced *Self-Preference Bias* (SPB), systematically favouring arguments that structurally or stylistically align with its own pre-trained generative patterns, regardless of objective accuracy. This is a well-documented failure mode in LLM-as-a-Judge frameworks, and is particularly acute when the judge and the agents share the same model family [1,7].

The Mitigation. To support fair adjudication in a zero-shot setting, the system will apply black-box perturbations (e.g., synonym replacement) to the Proposer’s and Critic’s outputs to obfuscate authorship. Furthermore, the Judge’s task will employ *cognitive load decomposition*, asking it to rate specific IRAC elements individually rather than issuing a single holistic score. This decomposition strategy mirrors the fine-grained, criterion-aware evaluation principles proposed in JudgeBoard [1].

7.2 Hallucination Loops and ReAct Syntax Failure

The Issue. SLMs notoriously fail to maintain the strict syntax required for ReAct (Thought-Action-Observation), causing the orchestrator to crash. Additionally, agents can fall into a *Hallucination Loop*, in which they hallucinate a tool output, validate their own hallucination, and confidently present a fabricated conclusion. Empirical studies confirm that this is a systemic weakness of SLMs in zero-shot function-calling scenarios [3].

The Mitigation. Strict schema enforcement via constrained sampling (Guidance/Outlines) guarantees that the SLM can only generate syntactically valid JSON/YAML. Invalid tokens are rejected at the compiler level before execution, physically preventing syntax-based hallucination loops. This approach is complemented by the resilient task-execution strategies developed in frameworks such as ResilioMate [8], which demonstrate the value of structured error-interception in SLM multi-agent pipelines.

7.3 Evaluating Active RAG and ReAct Tool Usage

The Issue. Evaluating a dynamic system requires moving beyond simple exact-match correctness to evaluating the *process* of retrieval and reasoning across multiple debate turns.

The Mitigation. Two complementary evaluation metrics are adopted.

- **Active RAG Evaluation:** Measured using the *Gain Beyond RAG* (GBR) metric [2], which quantifies the improvement in downstream accuracy specifically attributable to dynamic retrieval over a passive baseline. Search efficiency will also be tracked by penalising both over-searching and under-searching.
- **ReAct Evaluation:** Measured via *Turn Faithfulness*—ensuring tool outputs strictly ground the agent’s claims—and *Turn Contextual Precision*, evaluating the relevance of retrieved legal statutes at each debate turn.

8 System Features

The final engineered system will encapsulate the following six core features:

1. **Judge-Proposer-Critic Multi-Agent Topology:** Three distinct zero-shot agents orchestrated sequentially to debate legal queries, following the MAD paradigm of [4].
2. **LegalBench-RAG Integration:** A robust testing pipeline utilising a specialised 79M-character legal retrieval corpus.
3. **Grammar-Constrained Decoding:** Real-time token masking to enforce strict JSON ReAct syntax for tool execution, addressing the format-adherence failures documented in SLM function-calling studies [3].

4. **Adversarial Active RAG:** Dynamic tool-calling where the Critic and Proposer autonomously trigger vector database searches to ground their arguments or invalidate peers’, building on the GBR-trained search paradigm of [2] and the domain RAG advances of [9].
5. **Adaptive Stability Detection:** A mathematically driven stopping mechanism based on Kolmogorov-Smirnov testing over a Beta-Binomial distribution, to terminate debates optimally before social herding occurs [4].
6. **SPB-Mitigated Adjudication:** Obfuscation and cognitive decomposition prompting strategies to ensure the single 7B Judge remains impartial, informed by best practices from SLM judgment benchmarking [1,7].

9 Conclusion

This technical report has outlined a principled framework for scaling up the reasoning capabilities of Small Language Models through structured multi-agent debate, without incurring the computational and financial costs of large frontier models. By combining a Judge-Proposer-Critic topology with grammar-constrained decoding, adversarial Active RAG, and a mathematically grounded stopping criterion, the proposed system addresses the key failure modes of SLMs in zero-shot legal reasoning tasks. The framework synthesises a rich body of recent work on multi-agent orchestration [4,1,6,8,5,9], SLM structural limitations [3,7], and model-agnostic search optimisation [2]. Evaluation on LegalBench-RAG will provide a rigorous empirical test of whether emergent multi-agent consensus in small models can challenge the zero-shot supremacy of much larger parameter counts.

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